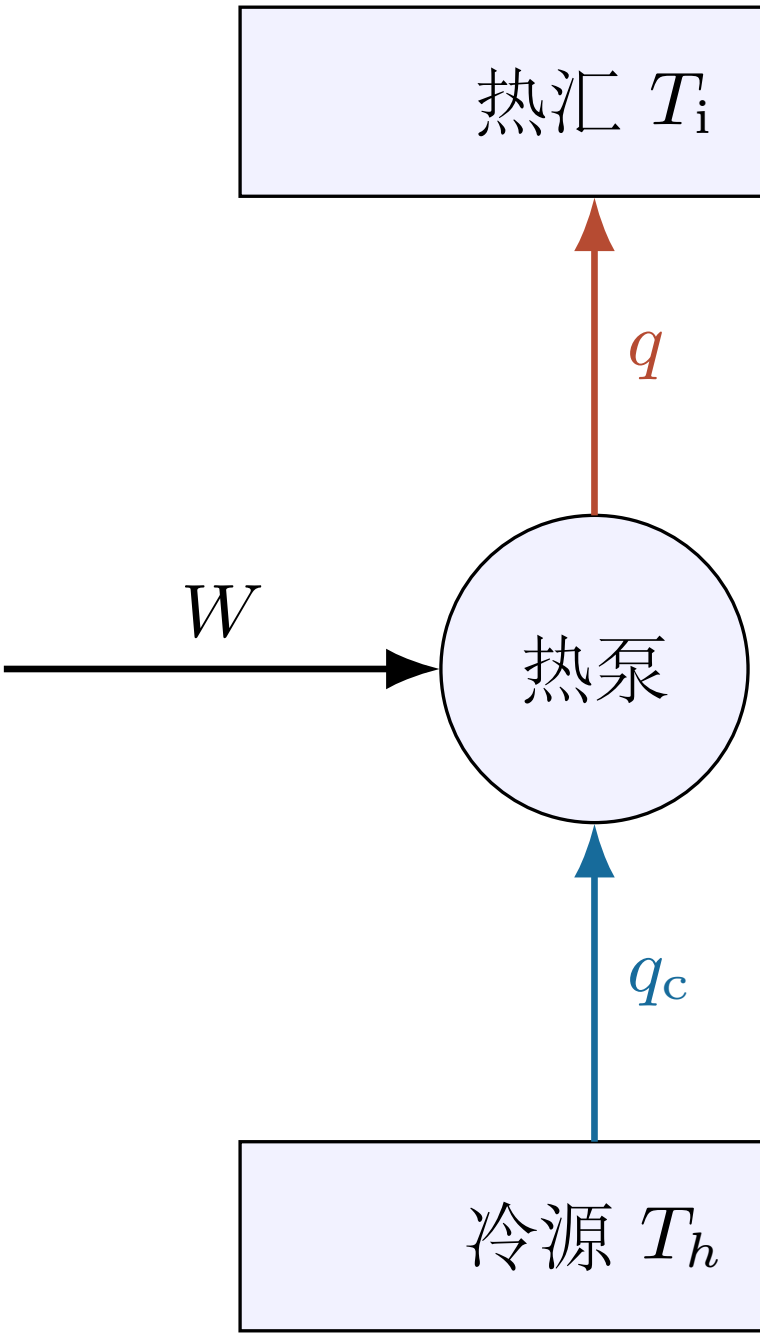

$$q = q_{\mathrm{c}} + W$$

$$\frac{q}{T_{\mathrm{i}}} \geq \frac{q_{\mathrm{c}}}{T_h}$$